

```

Last login: Tue Jun 16 18:54:05 on ttys000
psql
adityadave@Adityas-MacBook-Air-3 ~ % psql -d tpch
psql (18.4)
Type "help" for help.

tpch=# SET ROLE alice;
SET
tpch=> EXPLAIN (
ANALYZE true,
VERBOSE true,
COSTS true,
BUFFERS false,
TIMING false,
FORMAT JSON
)
select
  o_orderpriority,
  count(*) as order_count
from
  orders_secure
where
  o_orderdate >= date '1994-01-01'
and o_orderdate < date '1994-01-01' + interval '3' month
and exists (
  select
    *
  from
    lineitem
  where
    l_orderkey = o_orderkey
    and l_commitdate < l_receiptdate
)
group by
  o_orderpriority
order by
  o_orderpriority;
WARNING: ===== planner input query tree =====
DETAIL:
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE true
:isReturn false
:ctelist <>
:rtable (
  {RANGETBENTRY
:alias <>
:eref
  {ALIAS
:aliasname orders_secure
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
}
:rtekind 1
:subquery
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
  {RANGETBENTRY
:alias
  {ALIAS
:aliasname o
:colnames <>
}
}
:eref
  {ALIAS
:aliasname o
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
}
:rtekind 0
:relid 16561
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16561
:inh true
:requiredPerms 2
:checkasuser 10
:selectedCols (b 8 9 10 11 12 13 14 15 16)
:insertedCols (b)
:updatedCols (b)
}
)
}
:jointree
{FROMEXPR
:fromlist (
  {RANGETBLREF
:rtindex 1
}
)
}
:quals
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
  {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
}
}
}

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:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTY
    :alias
    {ALIAS
      :aliasname c
      :colnames <>
    }
    :eref
    {ALIAS
      :aliasname c
      :colnames ("c_custkey" "c_name" "c_address" "c_nationk
        ey" "c_phone" "c_acctbal" "c_mktsegment" "c_comment")
    }
    :rtekind 0
    :relid 16546
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 16546
    :inh true
    :requiredPerms 2
    :checkAsUser 10
    :selectedCols (b 8 13)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:joinres
{FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
  )
  :quals
  {BOOLEXP
    :boolop and
    :args (
      {OPEXP
        :opno 96
        :opfuncid 65
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {VAR
            :varno 1
            :varattno 1
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 1
            :location -1
          }
          {VAR
            :varno 1
            :varattno 2
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 1
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 2
            :location -1
          }
        )
      }
      :location -1
    }
    {OPEXP
      :opno 674
      :opfuncid 297
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {VAR
          :varno 1
          :varattno 6
          :vartype 701
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 6
          :location -1
        }
        {FUNCEXP
          :funcid 316
          :funcresulttype 701
          :funcset false
          :funcvariadic false
          :funcformat 2
          :funccollid 0
          :inputcollid 0
          :args (
            {CONST
              :consttype 23
              :consttypmod -1
              :constcollid 0
              :constlen 4
              :constbyval true
              :constisnull false
              :location -1
              :constvalue 4 [ -120 19 0 0 0 0 0 0 ]
            }
          )
        }
      }
      :location -1
    }
  )
  :location -1
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {CONST
      :consttype 23

```

```

        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgrouppref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
}
:resno 1
:resname o_orderkey
:ressortgrouppref 0
:resorigtbl 16561
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
}
:resno 2
:resname o_custkey
:ressortgrouppref 0
:resorigtbl 16561
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
}
:resno 3
:resname o_orderstatus
:ressortgrouppref 0
:resorigtbl 16561
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
}
:resno 4
:resname o_totalprice
:ressortgrouppref 0
:resorigtbl 16561
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
}

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:resno 5
:resname o_orderdate
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:varstype 25
:varstypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
}
:resno 6
:resname o_orderpriority
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:varstype 25
:varstypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
}
:resno 7
:resname o_clerk
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
}
:resno 8
:resname o_shippriority
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:varstype 25
:varstypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location -1
}
}
:resno 9
:resname o_comment
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 9
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier false
:relid 16956
:inh false
:relkind v
:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("o_orderpriority")
}
}
:rtckind 9
:groupexprs (
{VAR
:varno 1
:varattno 6
:varstype 25
:varstypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6

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        :location 116
    }
}
:lateral false
:inFromCl false
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16956
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 12 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointtree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:varstype 1082
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 194
}
{CONST
:consttype 1082
:conststypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 214
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
}
)
}
:location 206
}
{OPEXP
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:varstype 1082
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 235
}
{OPEXP
:opno 1076
:opfuncid 2071
:opresulttype 1114
:opretset false
:opcollid 0
:inputcollid 0
:args (
{CONST
:consttype 1082
:conststypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 254
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
}
{CONST
:consttype 1186
:conststypmod 196607
:constcollid 0
:constlen 16
:constbyval false
:constisnull false
:location 278
:constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 ]
}
)
}
:location 267
}
)
:location 247
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:opetName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStat <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTRY
:alias <>
:eref
{ALIAS

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:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenu
mber" "l_quantity" "l_extendedprice" "l_discount" "l_tax"
"l_returnflag" "l_linestatus" "l_shipdate" "l_commitdate
" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_commen
t")
}
:rtekind 0
:relid 16566
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
23)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 394
}
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 407
}
}
:location 405
}
{OPEXP
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 434
}
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 449
}
}
:location 447
}
}
:location 430
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 332
}
}
:resno 1
:resname l_orderkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr

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{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 332
}
:resno 2
:resname 1_partkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 332
}
:resno 3
:resname 1_supkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 332
}
:resno 4
:resname 1_linenummer
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 332
}
:resno 5
:resname 1_quantity
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 332
}
:resno 6
:resname 1_extendedprice
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location 332
}
:resno 7
:resname 1_discount
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location 332
}
:resno 8
:resname 1_tax
:ressortgroupref 0

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:resorigtbl 16566
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location 332
}
:resno 9
:resname l_returnflag
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location 332
}
:resno 10
:resname l_linestatus
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 332
}
:resno 11
:resname l_shipdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 332
}
:resno 12
:resname l_commitdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 332
}
:resno 13
:resname l_receiptdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location 332
}
:resno 14
:resname l_shipinstruct
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1

```

```

        :varattnosyn 15
        :location 332
    }
    :resno 15
    :resname l_shipmode
    :ressortgroupref 0
    :resorigtbl 16566
    :resorigcol 15
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 16
    :varstype 25
    :varstypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 16
    :location 332
    }
    :resno 16
    :resname l_comment
    :ressortgroupref 0
    :resorigtbl 16566
    :resorigcol 16
    :resjunk false
    }
    )
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDeps <>
    :withCheckOptions <>
    :stmt_location 0
    :stmt_len 0
    }
    :location 296
    }
    }
    :location 231
    }
    }
    :mergeActionList <>
    :mergeFtargetRelation 0
    :mergeJoinCondition <>
    :targetList (
    {TARGETENTRY
    :expr
    {VAR
    :varno 2
    :varattno 1
    :varstype 25
    :varstypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location -1
    }
    :resno 1
    :resname o_orderpriority
    :ressortgroupref 1
    :resorigtbl 16956
    :resorigcol 6
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {AGGREF
    :aggfnoid 2803
    :aggtype 20
    :aggcollid 0
    :inputcollid 0
    :aggtranstype 0
    :aggargtypes <>
    :aggdirectargs <>
    :args <>
    :aggorder <>
    :aggdistinct <>
    :aggfilter <>
    :aggstar true
    :aggvaradic false
    :aggkind 0
    :aggpresorted false
    :agglevelsup 0
    :aggssplit 0
    :aggno -1
    :aggtransno -1
    :location 137
    }
    :resno 2
    :resname order_count
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    )
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause (
    {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sorttop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
    }
    )
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause (
    {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sorttop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
    }
    )

```

```

)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

```

WARNING: pull\_up\_sublinks - File: src/backend/optimizer/prop/prepjointree.c, Relation: N/A, Explanation: Initiating pull-up of sublinks (ANY/EXISTS)

WARNING: pull\_up\_sublinks - File: src/backend/optimizer/prop/prepjointree.c, Relation: N/A, Explanation: Initiating pull-up of sublinks (ANY/EXISTS)

WARNING: ===== after pull up and before planning path list =====

```

DETAIL:  {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTF true
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname orders_secure
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 16956
:inh false
:relkind v
:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("o_orderpriority")
}
:rtekind 9
:groupexprs (
{VAR
:varno 4
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 6
:location -1
}
)
:lateral false
:inFromCl false
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 16566
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname o
:colnames <>
}
:eref
{ALIAS
:aliasname o
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 0
:relid 16561
:inh false
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "c_acctbal" "c_mktsegment" "c_comment")
}
:rtekind 0
:relid 16546
:inh false
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}

```

```

)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16956
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 12 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16561
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14 15 16)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16546
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 4
}
)
:quals <>
}
):rarg
{RANGETBLREF
:rtindex 5
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 5
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location -1
}
{VAR
:varno 4
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location -1
}
)
:location -1
}
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 5
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 6
:location -1
}
{CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 -120 -77 64 ]
}
)
:location -1
}
)
:alias <>
:rtindex 0
}
)
:quals (
{OPEXPR
:opno 1098
:opfuncid 1090
:opresulttype 16

```

```

:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 4
    :varattno 5
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 5
    :location -1
  }
  {CONST
    :consttype 1082
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location 214
    :constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
  }
)
:location 206
}
{OPEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 4
    :varattno 5
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 5
    :location -1
  }
  {CONST
    :consttype 1114
    :consttypmod -1
    :constcollid 0
    :constlen 8
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 8 [ 0 32 105 23 -25 90 -1 -1 ]
  }
)
:location 247
}
)
}
:rarg
{RANGETBLREF
:rtindex 3
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 3
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 1
    :location 394
  }
  {VAR
    :varno 4
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 1
    :location -1
  }
)
:location 405
}
{OPEXPR
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 3
    :varattno 12
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 12
    :location 434
  }
  {VAR
    :varno 3
    :varattno 13
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 13
    :location 449
  }
)
:location 447
}
)

```

```

        :alias <>
        :rtindex 0
    }
}
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
    :expr
    {VAR
    :varno 4
    :varattno 6
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 6
    :location -1
    }
    :resno 1
    :resname o_orderpriority
    :ressortgroupref 1
    :resorigtbl 16956
    :resorigcol 6
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {AGGREF
    :aggfnoid 2803
    :aggtype 28
    :aggcollid 0
    :inputcollid 0
    :aggtranstype 0
    :aggargtypes <>
    :aggdirectargs <>
    :args <>
    :aggorder <>
    :aggdistinct <>
    :aggfilter <>
    :aggstar true
    :aggvariadic false
    :aggkind n
    :aggpresorted false
    :agglevelsup 0
    :aggsplit 0
    :aggnosyn -1
    :aggtransno -1
    :location 137
    }
    :resno 2
    :resname order_count
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
    {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
    }
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
    {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
    }
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

```

WARNING: ===== Sorted Qual List in order\_qual\_clauses =====

```

DETAIL: (
{RESTRICTINFO
:clause
{OPEXPR
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 5
:location -1
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 214
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
}
}
:location 206
}
:is_pushed_down true
:can_join false
:pseudoconstant false
:has_clone false
:is_clone false

```

```

:leakproof false
:has_volatile 0
:security_level 0
:num_base_rels 1
:clause_relds (b 4)
:required_relds (b 4)
:incompatible_relds (b)
:outer_relds (b)
:left_relds (b 4)
:right_relds (b)
:orclause <>
:rinfo_serial 3
:eval_cost.startup 0
:eval_cost.per_tuple 0.0025
:norm_selec 0.6922656734118627
:outer_selec -1
:mergeopfamilies <>
:left_em <>
:right_em <>
:outer_is_left false
:hashjoinoperator 0
:left_bucketsize -1
:right_bucketsize -1
:left_mcvfreq -1
:right_mcvfreq -1
:left_hasheqoperator 0
:right_hasheqoperator 0
}
{RESTRUCTINFO
:clause
{OEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 5
:location -1
}
{CONST
:consttype 1114
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 32 105 23 -25 90 -1 -1 ]
}
)
:location 247
}
}
:is_pushed_down true
:can_join false
:pseudoconstant false
:has_clone false
:is_clone false
:leakproof false
:has_volatile 0
:security_level 0
:num_base_rels 1
:clause_relds (b 4)
:required_relds (b 4)
:incompatible_relds (b)
:outer_relds (b)
:left_relds (b 4)
:right_relds (b)
:orclause <>
:rinfo_serial 4
:eval_cost.startup 0
:eval_cost.per_tuple 0.0025
:norm_selec 0.36386992886303763
:outer_selec -1
:mergeopfamilies <>
:left_em <>
:right_em <>
:outer_is_left false
:hashjoinoperator 0
:left_bucketsize -1
:right_bucketsize -1
:left_mcvfreq -1
:right_mcvfreq -1
:left_hasheqoperator 0
:right_hasheqoperator 0
}
}
)

```

WARNING: ===== create\_seqscan\_plan scan->plan.qual =====

```

DETAIL: (
{OEXPR
:opno 1090
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 5
:location -1
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 214
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
}
)
:location 206
}
{OEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 5
:vartype 1082
:vartypmod -1

```

```

:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 5
:location -1
}
{CONST
:consttype 1114
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 32 105 23 -25 90 -1 -1 ]
}
)
:location 247
}
)

```

WARNING: ===== create\_seqscan\_plan scan->plan.qual =====

```

DETAIL: (
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 5
:varattno 6
:vartype 781
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 6
:location -1
}
{CONST
:consttype 781
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 -120 -77 64 ]
}
)
:location -1
}
)

```

WARNING: ===== create\_seqscan\_plan scan->plan.qual =====

```

DETAIL: (
{OPEXPR
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 12
:location 434
}
{VAR
:varno 3
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 13
:location 449
}
)
:location 447
}
)

```

WARNING: === PLANNER SUGGESTION ===

```

Plan:
Finalize GroupAggregate (cost=194594.52..194617.59 rows=5 width=17)
Output: o.o_orderpriority, count(*)
Group Key: o.o_orderpriority
-> Gather Merge (cost=194594.52..194617.48 rows=12 width=17)
Output: o.o_orderpriority, (PARTIAL count(*))
Workers Planned: 2
-> Partial GroupAggregate (cost=193594.49..193616.07 rows=5 width=17)
Output: o.o_orderpriority, PARTIAL count(*)
Group Key: o.o_orderpriority
-> Sort (cost=193594.49..193601.67 rows=2870 width=9)
Output: o.o_orderpriority
Sort Key: o.o_orderpriority
-> Parallel Hash Semi Join (cost=154455.47..193429.66 rows=2870 width=9)
Output: o.o_orderpriority
Hash Cond: (o.o_orderkey = lineitem.l_orderkey)
-> Parallel Hash Semi Join (cost=4806.41..40362.28 rows=10210 width=13)
Output: o.o_orderpriority, o.o_orderkey
Hash Cond: (o.o_custkey = c.c_custkey)
-> Parallel Seq Scan on public.orders o (cost=0.00..35383.00 rows=22585 width=17)
Output: o.o_orderkey, o.o_custkey, o.o_orderstatus, o.o_totalprice, o.o_orderdate, o.o_orderpriority, o.o_clerk, o.o_shippriority, o.o_comment
Filter: ((o.o_orderdate >= '1994-01-01':date) AND (o.o_orderdate < '1994-04-01 00:00:00':timestamp without time zone))
-> Parallel Hash (cost=4453.25..4453.25 rows=28253 width=4)
Output: c.c_custkey
-> Parallel Seq Scan on public.customer c (cost=0.00..4453.25 rows=28253 width=4)
Output: c.c_custkey
Filter: (c.c_acctbal > '5000':double precision)
-> Parallel Hash (cost=135974.79..135974.79 rows=833461 width=4)
Output: lineitem.l_orderkey
-> Parallel Seq Scan on public.lineitem (cost=0.00..135974.79 rows=833461 width=4)
Output: lineitem.l_orderkey
Filter: (lineitem.l_commitdate < lineitem.l_receiptdate)

```

WARNING: === EXECUTOR EXECUTION ===

```

Rows Processed: 5
Plan & Stats:
Finalize GroupAggregate (cost=194594.52..194617.59 rows=5 width=17) (actual time=1346.223..1372.311 rows=5.00 loops=1)
Output: o.o_orderpriority, count(*)
Group Key: o.o_orderpriority
Buffers: shared hit=4440 read=129976, temp read=20152 written=20540
-> Gather Merge (cost=194594.52..194617.48 rows=12 width=17) (actual time=1346.045..1372.300 rows=15.00 loops=1)
Output: o.o_orderpriority, (PARTIAL count(*))
Workers Planned: 2
Workers Launched: 2
Buffers: shared hit=4440 read=129976, temp read=20152 written=20540

```

-> Partial GroupAggregate (cost=193594.49..193616.07 rows=5 width=17) (actual time=1340.295..1340.967 rows=5.00 loops=3)  
Output: o.o\_orderpriority, PARTIAL count(\*)  
Group Key: o.o\_orderpriority  
Buffers: shared hit=4440 read=129976, temp read=20152 written=20540  
Worker 0: actual time=1337.467..1338.037 rows=5.00 loops=1  
Buffers: shared hit=1484 read=43144, temp read=6736 written=6908  
Worker 1: actual time=1337.885..1338.499 rows=5.00 loops=1  
Buffers: shared hit=1495 read=43149, temp read=6742 written=6848  
-> Sort (cost=193594.49..193601.67 rows=2870 width=9) (actual time=1340.136..1340.478 rows=7772.33 loops=3)  
Output: o.o\_orderpriority  
Sort Key: o.o\_orderpriority  
Sort Method: quicksort Memory: 385kB  
Buffers: shared hit=4440 read=129976, temp read=20152 written=20540  
Worker 0: actual time=1337.318..1337.580 rows=7360.00 loops=1  
Sort Method: quicksort Memory: 193kB  
Buffers: shared hit=1484 read=43144, temp read=6736 written=6908  
Worker 1: actual time=1337.731..1338.813 rows=7727.00 loops=1  
Sort Method: quicksort Memory: 193kB  
Buffers: shared hit=1495 read=43149, temp read=6742 written=6848  
-> Parallel Hash Semi Join (cost=154455.47..193429.66 rows=2870 width=9) (actual time=1210.308..1338.029 rows=7772.33 loops=3)  
Output: o.o\_orderpriority  
Hash Cond: (o.o\_orderkey = lineitem.l\_orderkey)  
Buffers: shared hit=4424 read=129976, temp read=20152 written=20540  
Worker 0: actual time=1211.402..1335.509 rows=7360.00 loops=1  
Buffers: shared hit=1476 read=43144, temp read=6736 written=6908  
Worker 1: actual time=1211.383..1335.459 rows=7727.00 loops=1  
Buffers: shared hit=1487 read=43149, temp read=6742 written=6848  
-> Parallel Hash Semi Join (cost=4806.41..40362.28 rows=10210 width=13) (actual time=15.152..159.631 rows=8461.00 loops=3)  
Output: o.o\_orderpriority, o.o\_orderkey  
Hash Cond: (o.o\_custkey = c.c\_custkey)  
Buffers: shared hit=4048 read=25632  
Worker 0: actual time=15.066..159.500 rows=8461.00 loops=1  
Buffers: shared hit=1353 read=8512  
Worker 1: actual time=15.276..159.517 rows=8484.00 loops=1  
Buffers: shared hit=1361 read=8569  
-> Parallel Seq Scan on public.orders o (cost=0.00..35383.00 rows=22585 width=17) (actual time=0.481..138.005 rows=18783.00 loops=3)  
Output: o.o\_orderkey, o.o\_custkey, o.o\_orderstatus, o.o\_totalprice, o.o\_orderdate, o.o\_orderpriority, o.o\_clerk, o.o\_shippriority, o.o\_comment  
Filter: ((o.o\_orderdate >= '1994-01-01'::date) AND (o.o\_orderdate < '1994-04-01 00:00:00'::timestamp without time zone))  
Rows Removed by Filter: 481217  
Buffers: shared hit=376 read=25632  
Worker 0: actual time=0.478..137.920 rows=18876.00 loops=1  
Buffers: shared hit=131 read=8512  
Worker 1: actual time=0.692..137.980 rows=18849.00 loops=1  
Buffers: shared hit=131 read=8569  
-> Parallel Hash (cost=4453.25..4453.25 rows=28253 width=4) (actual time=14.459..14.459 rows=22663.00 loops=3)  
Output: c.c\_custkey  
Buckets: 131072 Batches: 1 Memory Usage: 3712kB  
Buffers: shared hit=3672  
Worker 0: actual time=14.544..14.544 rows=22497.00 loops=1  
Buffers: shared hit=1222  
Worker 1: actual time=14.526..14.527 rows=22653.00 loops=1  
Buffers: shared hit=1230  
-> Parallel Seq Scan on public.customer c (cost=0.00..4453.25 rows=28253 width=4) (actual time=0.022..11.084 rows=22663.00 loops=3)  
Output: c.c\_custkey  
Filter: (c.c\_acctbal > '5000'::double precision)  
Rows Removed by Filter: 27337  
Buffers: shared hit=3672  
Worker 0: actual time=0.023..11.075 rows=22497.00 loops=1  
Buffers: shared hit=1222  
Worker 1: actual time=0.021..11.066 rows=22653.00 loops=1  
Buffers: shared hit=1230  
-> Parallel Hash (cost=135974.79..135974.79 rows=833461 width=4) (actual time=1027.552..1027.552 rows=1264432.00 loops=3)  
Output: lineitem.l\_orderkey  
Buckets: 262144 (originally 262144) Batches: 32 (originally 16) Memory Usage: 6784KB  
Buffers: shared hit=376 read=104344, temp read=9149 written=20156  
Worker 0: actual time=1025.385..1025.386 rows=1258964.00 loops=1  
Buffers: shared hit=123 read=34632, temp read=3126 written=6780  
Worker 1: actual time=1025.378..1025.378 rows=1257324.00 loops=1  
Buffers: shared hit=126 read=34580, temp read=3008 written=6720  
-> Parallel Seq Scan on public.lineitem (cost=0.00..135974.79 rows=833461 width=4) (actual time=0.776..535.737 rows=1264432.00 loops=3)  
Output: lineitem.l\_orderkey  
Filter: (lineitem.l\_commitdate < lineitem.l\_receiptdate)  
Rows Removed by Filter: 735973  
Buffers: shared hit=376 read=104344  
Worker 0: actual time=0.532..534.183 rows=1258964.00 loops=1  
Buffers: shared hit=123 read=34632  
Worker 1: actual time=0.806..535.720 rows=1257324.00 loops=1  
Buffers: shared hit=126 read=34580

QUERY PLAN

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          "Parent Relationship": "Outer",
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          "Total Cost": 194617.48,
          "Plan Rows": 12,
          "Plan Width": 17,
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          "Actual Loops": 1,
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              "Async Capable": false,
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              "Total Cost": 193616.07,
              "Plan Rows": 5,
              "Plan Width": 17,
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              "Actual Loops": 3,
              "Disabled": false,
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              ]
            }
          ]
        },
        {
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```

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"Startup Cost": 193594.49,
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    "Async Capable": false,
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        "Actual Loops": 1
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    "Inner Unique": false,
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        "Actual Loops": 1
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        "Actual Loops": 1
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    "Async Capable": false,
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    "Plan Width": 17,
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    "Actual Loops": 3,
    "Disabled": false,
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    "Filter": "((o.o_orderdate >= '1994-01-01':date) AND (o.o_orderdate < '1994-04-01 00:00:00':timestamp without time zone))",
    "Rows Removed by Filter": 481217,
    "Workers": [
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        "Actual Loops": 1
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        "Actual Loops": 1
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    ]
  },
  {
    "Node Type": "Hash",
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    "Async Capable": false,
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    "Workers": [
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        "Actual Rows": 22497.00,
        "Actual Loops": 1
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        "Worker Number": 1,
        "Actual Rows": 22653.00,

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